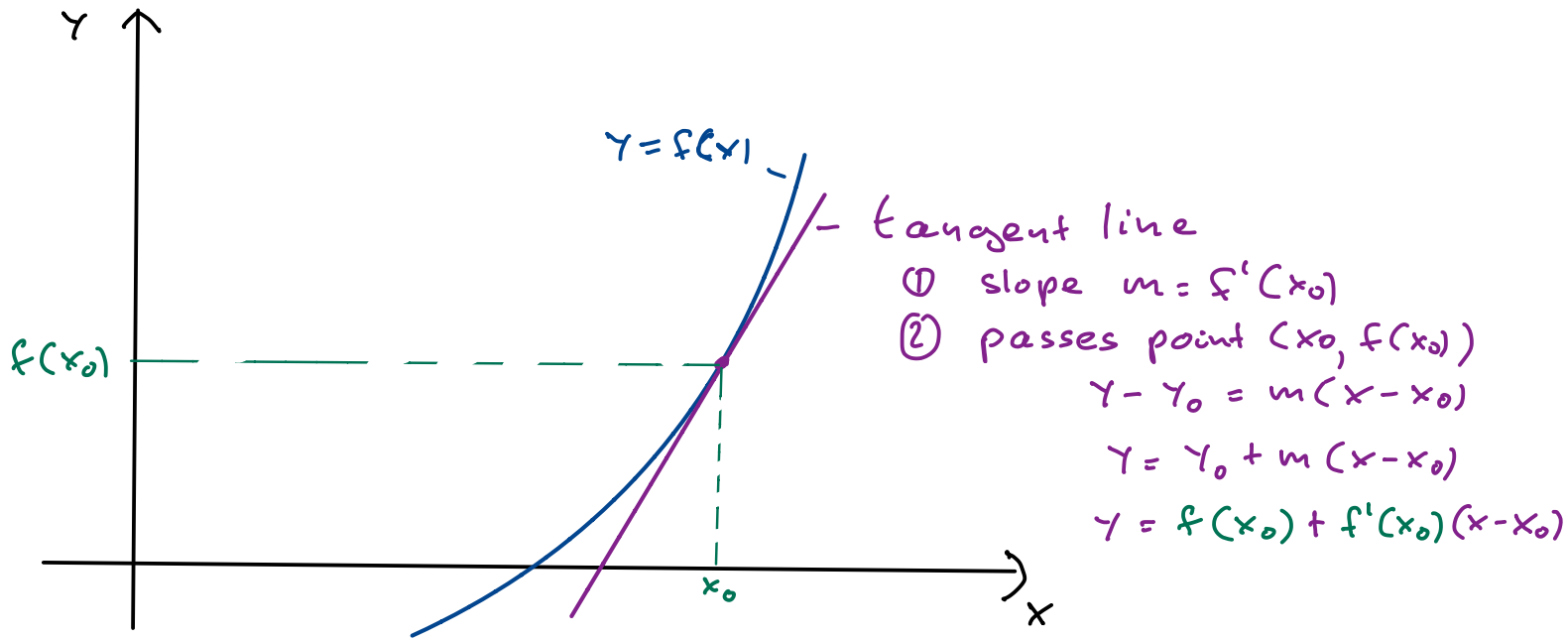
V.3 Newton's Method

Recall from Calculus I: Let the function $f(x)$ be differentiable at x_o.

Then the equation of the tangent line at the point $\big(x_o,\, f(x_o)\big)$ is

$$\boxed{y \;=\; p_1(x) \;=\; f(x_o) + f'(x_o)(x - x_o).}$$



which is the Taylor polynomial of degree one at x_o.

Task: Solve $f(x) = 0$

Newton's method is an iterative method for finding a root of the function $f(x)$ that is continuously differentiable. It is based on the following idea:

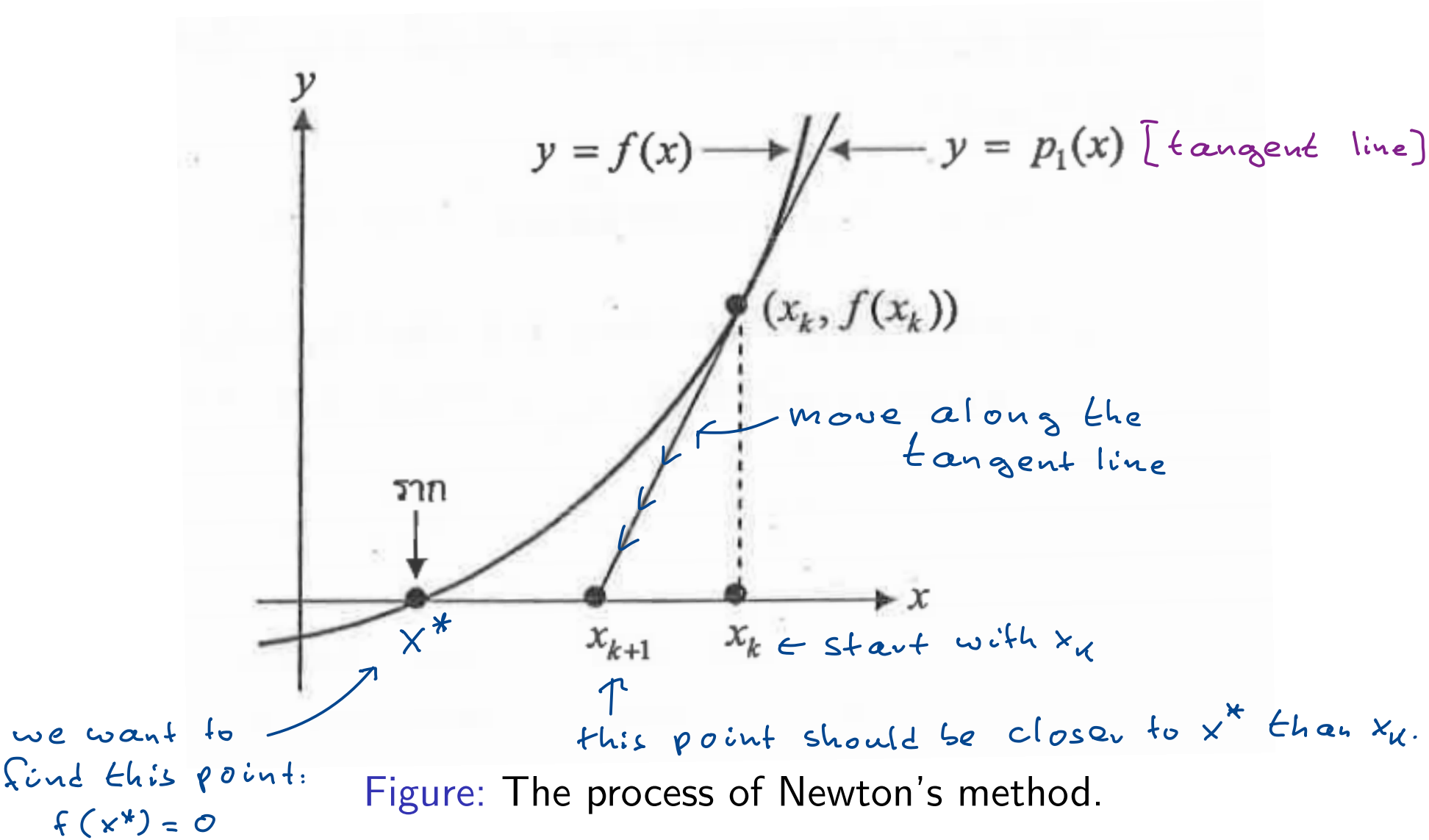


Figure: The process of Newton's method.

1. Let $\big(x_k, f(x_k)\big)$ be a point on the graph of f.
2. Construct the tangent line to the graph of f at the point $\big(x_k, f(x_k)\big)$:

$$\boxed{y \;=\; p_1(x) \;=\; f(x_k) + f'(x_k)(x - x_k)} \tag{3}$$

3. Let x_{k+1} be the intersection of this tangent line with the x-axis.

   (That is, x_{k+1} is the root of $p_1(x)$.)

   To find x_{k+1}, we solve equation (3) with $x = x_{k+1}$:

$$Y = 0 \Rightarrow \qquad f(x_k) + f'(x_k)(x_{k+1} - x_k) = 0 \tag{4}$$

$$f'(x_k)(x_{k+1} - x_k) = -f(x_k)$$

$$x_{k+1} - x_k = -\frac{f(x_k)}{f'(x_k)}$$

   so that

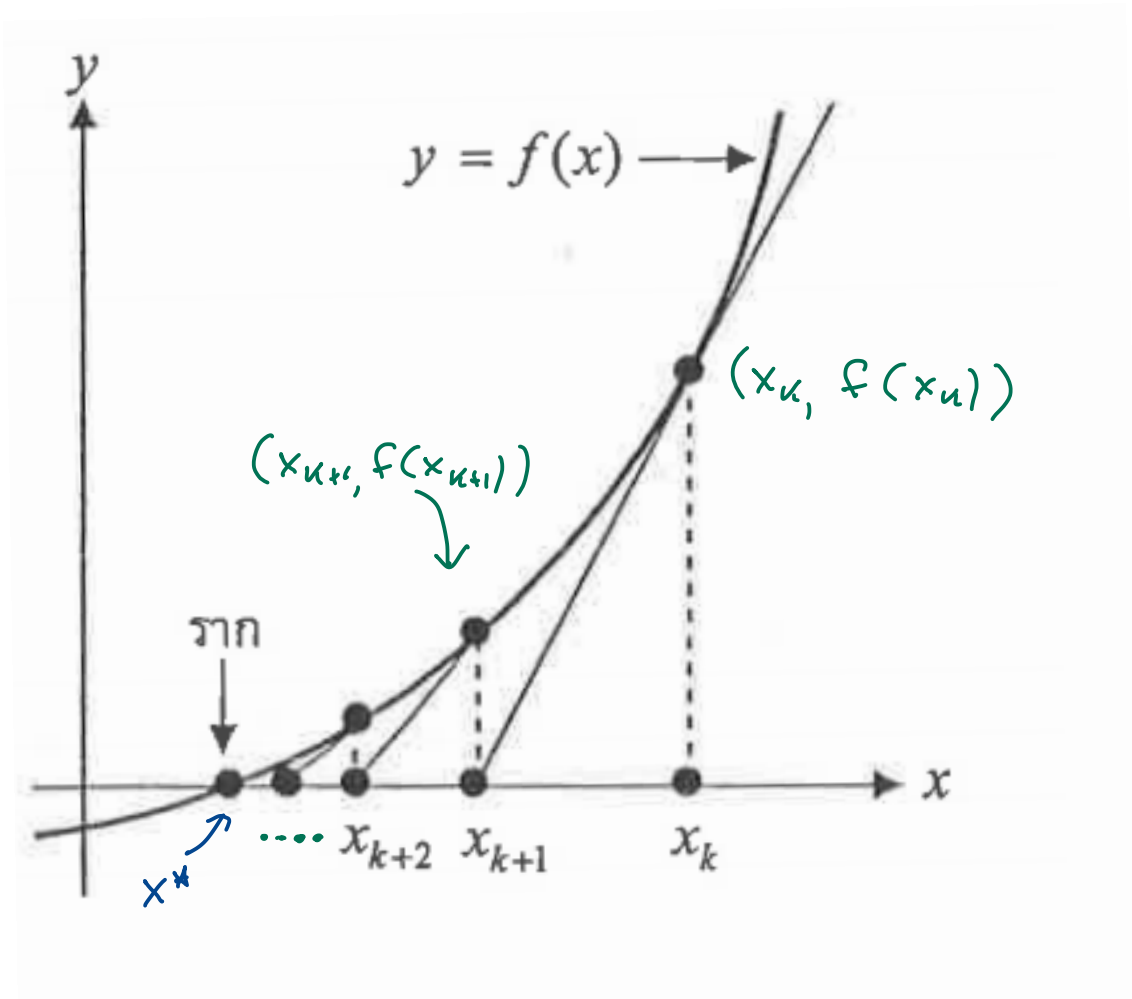
$$\boxed{x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}} \qquad (\text{if } f'(x_k) \neq 0). \tag{5}$$

   This is the formula for the Newton (iterative) method.

One first guesses an initial value x_0. Newton's formula will then compute x_1, and from x_1, will compute x_2, and so on. One thus obtains a sequence

$$\{x_0, x_1, x_2, \ldots, x_k, x_{k+1}, \ldots\}.$$

With luck this sequence will converge to a root of $f(x)$:



Common stopping conditions: One can stop the calculations when

1. the absolute difference between x_k and x_{k+1} is small:

$$|x_{k+1} - x_k| < \epsilon,$$

2. or the relative difference between x_k and x_{k+1} is small:

$$\left| \frac{x_{k+1} - x_k}{x_{k+1}} \right| < \epsilon,$$

3. or when $f(x_k)$ is close to zero,

$$|f(x_k)| < \epsilon$$

where ϵ is the accepted error.

Example 3: Use Newton's method to estimate $\sqrt{2}$

$x^* = \sqrt{2}$ is the positive solution of $x^2 - 2 = 0$

Solution: Set $f(x) = x^2 - 2$. Then $f'(x) = 2x$.

Newton's formula
$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

becomes
$$x_{k+1} = x_k - \frac{x_k^2 - 2}{2x_k}.$$

With $x_0 = 1$, we get: [choose x_0 close to x^*]

$k=0$: $x_1 = x_0 - \frac{x_0^2 - 2}{2x_0} = 1 - \frac{1^2 - 2}{2 \cdot 1} = 1 + \frac{1}{2} = 1.5$

$k=1$: $x_2 = x_1 - \frac{x_1^2 - 2}{2 \cdot x_1} = 1.5 - \frac{1.5^2 - 2}{2 \cdot 1.5} = 1.416667$

$k=2$: $x_3 = 1.414215$ $\left[x_3 = x_2 - \frac{x_2^2 - 2}{2 \cdot x_2}\right]$

$k=3$: $x_4 = 1.414214$ $\left[x_4 = x_3 - \frac{x_3^2 - 2}{2 \cdot x_3}\right]$

Since $|x_4 - x_3| \leq 10^{-6}$ we stop. Our estimated root is $x^\star = \underline{1.414214}$.

We notice that this method is faster than the bisection method.

Example 4: Use Newton's method to estimate the root of $f(x) = e^{-x} - x$ by letting $x_0 = 0$.

We solve: $e^{-x} - x = 0$

$\boxed{e^{-x} = x}$

Solution: The derivative is $f'(x) = -e^{-x} - 1$.

Newton's formula $$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

becomes $$x_{k+1} = x_k - \frac{e^{-x_k} - x_k}{-e^{-x_k} - 1}.$$ $\leftarrow f(x_k)$ $\leftarrow f'(x_k)$

With $x_0 = 0$, we get:

$$x_1 = 0.500000$$

$$x_2 = 0.566311$$

$$x_3 = 0.567143$$

$$x_4 = 0.567143$$

$k=0$: $x_1 = x_0 - \frac{e^{-x_0} - x_0}{-e^{-x_0} - 1}$

$= 0 - \frac{e^{-0} - 0}{-e^{-0} - 1} = -\frac{1-0}{-1-1} = \frac{1}{2} = 0.5$

$k=1$: $x_2 = x_1 - \frac{e^{-x_1} - x_1}{-e^{-x_1} - 1} = 0.5 - \frac{e^{-0.5} - 0.5}{-e^{-0.5} - 1}$

$= \cdots = 0.566311$

Since $x_4 = x_3$ we stop. Our estimated root is $x^\star = 0.567143$.

Exercise: Use Newton's method to find the point of intersection of $g(t) = 4t^2 + 2$ and $h(t) = 0.5t^3 + 6t$.
Hint: find the root of, say, $h(t) - g(t)$.

Exercise: Formulate Newton's method to

1. Compute the square and cube roots of 3.
2. Compute the nth root of a.
3. Compute the reciprocal of a.

Exercise: Solve the following equations using Newton's method by finding positive solutions. Arguments of trigonometric functions are in radians.

1. $\sin x = \frac{x}{2}$
2. $x^3 = \sin x + 7$
3. $\ln\left(x^2\right) = 0.5$
4. $e^x = x - 0.8$

Exercise: Use Newton's method to find a root of $f(x) = -0.9x^2 + 1.7x + 2.5$ with initial value $x_0 = 5.0$. Stop computing when the relative error is less than 1% by comparing with the true value of the root.
Hint: Use the quadratic formula to find the true value of the solution.

Exercise: Use Newton's method to find the *smallest* positive root of

$$f(x) = \sin x + \cos\left(1 + x^2\right) - 1$$

when x is in radians. Use 4 iterations with the following initial values:

1. $x_0 = 1.0$
2. $x_0 = 1.5$
3. $x_0 = 2.5$
4. $x_0 = 3.0$

You may use Python for example to sketch the graph of $f(x)$ to verify.

Error analysis. Let $x^\star$ be a root of $f(x)$, i.e. $f(x^\star) = 0$. Equation (4),

$$p_1(x_{k+1}) = f(x_k) + f'(x_k)(x_{k+1} - x_k) = 0 \tag{6}$$

shows that x_{k+1} is the root of the Taylor polynomial $p_1(x)$ of $f(x)$ centered at x_k . Adding the remainder term to $p_1(x)$,

$$f(x) = f(x_k) + f'(x_k)(x - x_k) + \underbrace{\frac{f''(\xi)}{2!}(x - x_k)^2}_{\text{Remainder term.}}$$

for all x. When $x = x^\star$ this becomes

$x^* =$ solution of $f(x)=0$

$$f(x^\star) = f(x_k) + f'(x_k)(x^\star - x_k) + \frac{f''(\xi)}{2!}(x^\star - x_k)^2 = 0 \tag{7}$$

ξ is between x_k and x^*

Now subtract equation (6) from equation (7) to get

$$f'(x_k)\underbrace{(x^\star - x_{k+1})}_{\text{Call this } E_{k+1}} + \frac{f''(\xi)}{2}\underbrace{(x^\star - x_k)}_{\text{Call this } E_k}{}^2 = 0, \tag{8}$$

where ξ lies between $x^\star$ and x_k.

errors of approximation

Let $E_k = x^\star - x_k$, $E_{k+1} = x^\star - x_{k+1}$ be the errors in iterations k and $k+1$ respectively. We solve (8) for E_{k+1}:

when
① x_0 is close to x^*
② $f'(x^*) \neq 0$
then this is nearly constant

$$E_{k+1} = \boxed{-\frac{f''(\xi)}{2f'(x_k)}} E_k^2. \tag{9}$$

ξ is between x^* and x_k

This shows that the error in iteration $k+1$ varies according to the square of the error in interation k; we call this **quadratic convergence**, which is considered very fast in practice.

Remark: The convergence of Newton's method depends on the initial value x_0: Only if x_0 is close enough to the root $x^\star$, then E_0 will be of small value. E_1 will be even smaller followed by an even smaller value of E_2, etc. For this reason, Newton's method is classified as a **local method**.

Newton's method, if it converges to a root, converges fast.
But it does not always converge:

The figure shows an x_0 that is quite far from the root and results in values for $x_1, x_2, x_3, \ldots$ moving away from $x^\star$, and thus fail to converge to the root.

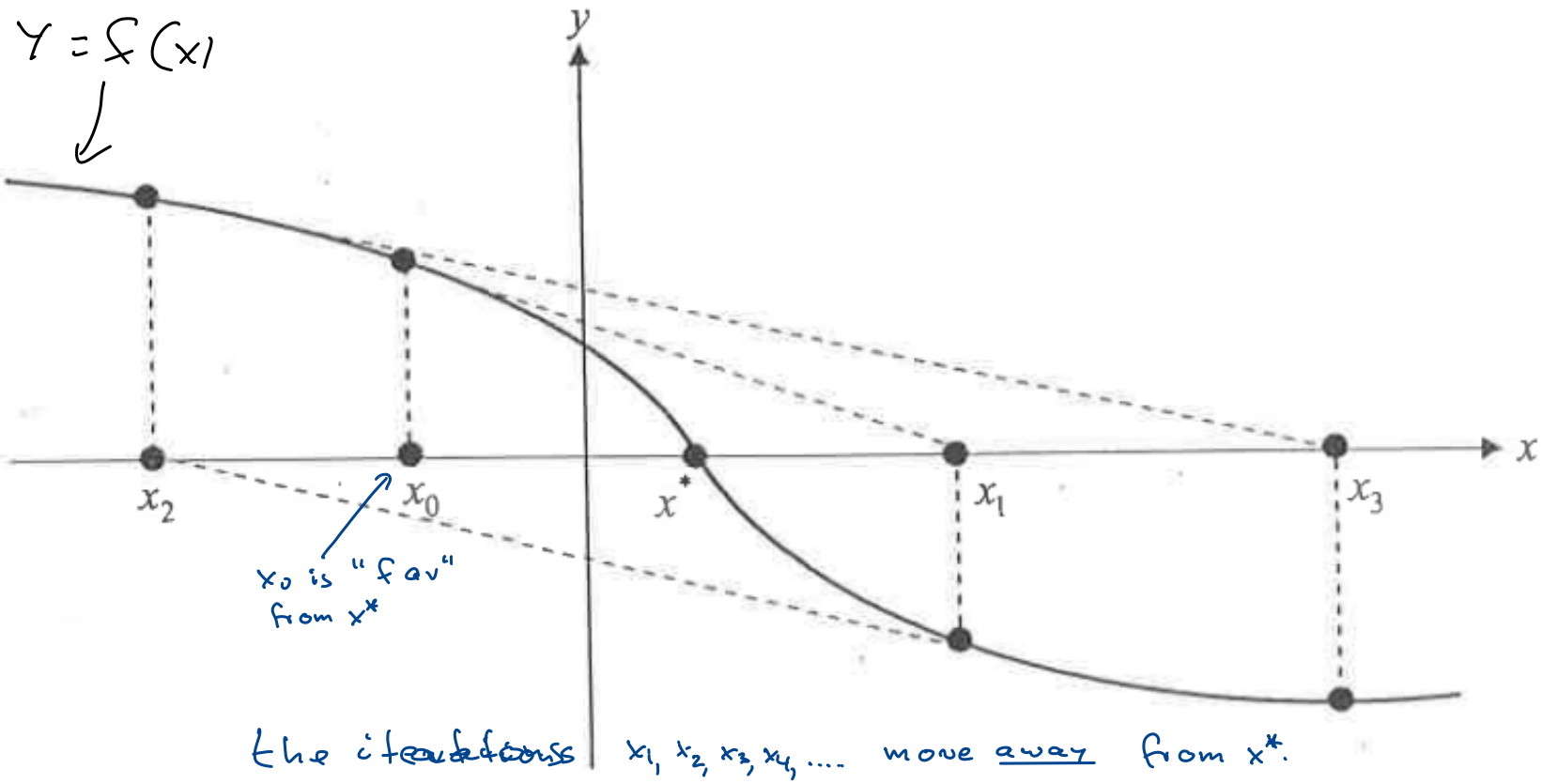


Figure: Newton's method fails to converge: a special case.

V.4 The Secant Method

Newton's method for finding a root of $f(x)$ requires computing the derivative $f'(x)$ at each iteration which, in more complicated problems, may use up significant computer resources.

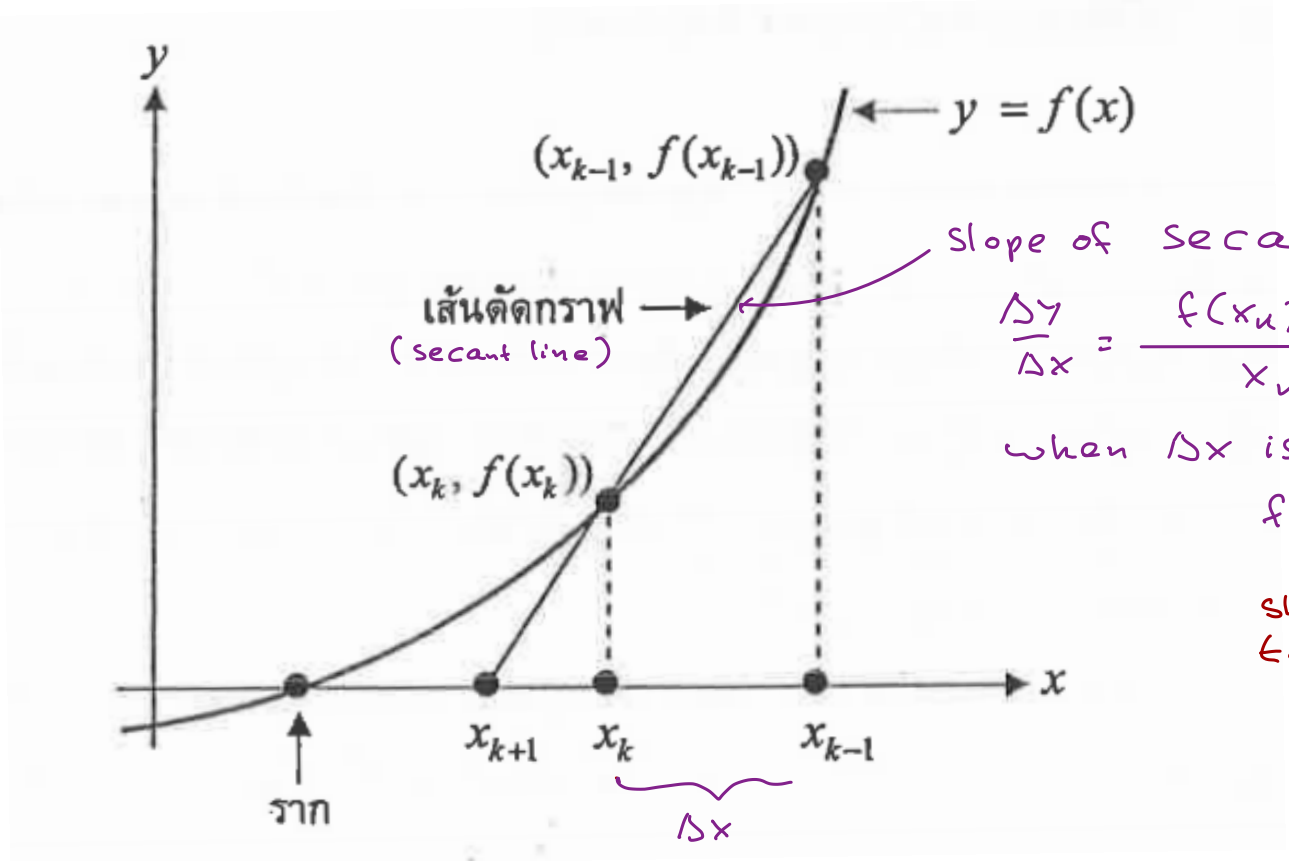
There exist, however, methods for finding the root of $f(x)$ that are based on Newton's method, but involve estimates of the derivative $f'(x)$. These are called **quasi Newton methods**.

One of these is the secant method.

From Newton's formula, $$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \tag{10}$$

we estimate $f'(x_k)$ by the slope of the **secant line** passing through the points $(x_k, f(x_k))$ and $(x_{k-1}, f(x_{k-1}))$:

$$f'(x_k) \approx \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$$



slope of secant line :

$$\frac{\Delta y}{\Delta x} = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} = m$$

when Δx is small, then

$$f'(x_k) \approx m$$

slope of tangent line — slope of secant line

The corresponding modification of equation (10) gives us a new formula:

$$\boxed{x_{k+1} = x_k - \frac{(x_k - x_{k-1})f(x_k)}{f(x_k) - f(x_{k-1})}}. \tag{11}$$

This quasi Newton method is called the **secant method**.

The first iteration of the secant method requires two values: x_0 and x_1; then x_2 is found. After that x_1 and x_2 are used to compute x_3, and so on. In general, x_{k-1} and x_k are used to compute x_{k+1}.

The stopping citeria are the same as with the Newton's method.

Example 5: Use the secant method to find $\sqrt{2}$.

Solution: Set $f(x) = x^2 - 2$. again. [We need not compute $f'(x)$]

We start with $x_0 = 1$ and $x_1 = 1.2$.

$$x_{k+1} = x_k - \frac{(x_k - x_{k-1}) f(x_k)}{f(x_k) - f(x_{k-1})}$$

k	x_k	$f(x_k)$
0	$x_0 =$ 1	-1
1	$x_1 =$ 1.2	-0.56
2	1.454545	0.115701
3	1.410959	-0.009195
4	1.414168	-0.000129
5	1.414214	0.000001

$k=1$: $x_2 = x_1 - \frac{(x_1 - x_0) f(x_1)}{f(x_1) - f(x_0)}$

$= 1.2 - \frac{(1.2-1)\cdot(-0.56)}{-0.56-(-1)}$

$= 1.2 - \frac{0.2\cdot(-0.56)}{0.44}$

$= 1.2 - \frac{-0.112}{0.44} = 1.454545$

$k=2$: $x_3 = x_2 - \frac{(x_2 - x_1) f(x_2)}{f(x_2) - f(x_1)}$

$= 1.454545 - \frac{(1.454545 - 1.2)\cdot(0.115701)}{0.115701 - (-0.56)}$

$= 1.410959$

Calculator

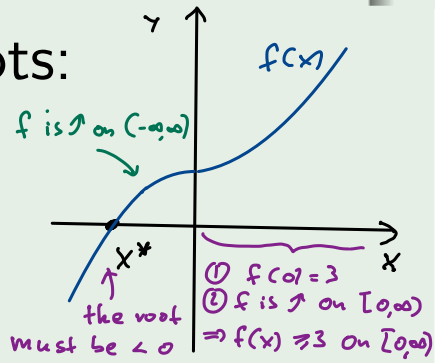
.....

Therefore, $\sqrt{2} \approx 1.414214$.

Close to 0.

Example 6: Use the secant method to find the largest real root of $f(x) = x^5 + x^3 + 3$.

Solution: First analyze the number and location of the roots:



1. $\lim_{x\to-\infty} f(x) = -\infty$, $\lim_{x\to\infty} f(x) = \infty$.

   Therefore, $f(x)$ has at least one root.

2. Note that $f'(x) = 5x^4 + 3x^2 \geq 0$, and $f'(x) = 0$ at $x = 0$ only.

   so that $f(x)$ is strictly increasing on $(-\infty, \infty)$.

   Therefore, $f(x)$ has at most one root.

   By 1. and 2. $f(x)$ has exactly one root.

3. By 2. $f(x) > 0$ for $x \geq 0$. Therefore, his root must be negative. $\Rightarrow x^* < 0$

Next we choose $x_0 = 1$ and $x_1 = -1$ and use the secant formula:

$$x_{k+1} = x_k - \frac{(x_k - x_{k-1})f(x_k)}{f(x_k) - f(x_{k-1})}$$

$f(x) = x^5 + x^3 + 3$

k	x_k	$f(x_k)$
0	$x_0 = 1$	$f(1) = 5$
1	$x_1 = -1$	$f(-1) = 1$
2	$x_2 = -1.5$	-7.96875
3	$x_3 = -1.05575$	0.51164
4	$x_4 = -1.08255$	
$\vdots$	$\vdots$	$\vdots$
7	-1.10530	4.768×10^{-7}
8	-1.10530	4.768×10^{-7}

$k=1$: $x_2 = x_1 - \frac{(x_1 - x_0) f(x_1)}{f(x_1) - f(x_0)}$

$= -1 - \frac{(-1-1)\cdot(1)}{1-5}$

$= -1 - \frac{-2}{-4} = -1 - \frac{1}{2} = -1.5$

$k=2$: $x_3 = x_2 - \frac{(x_2 - x_1) f(x_2)}{f(x_2) - f(x_1)}$

$= -1.5 - \frac{[-1.5-(-1)](-7.96875)}{-7.96875 - 1}$

$= -1.5 - \frac{(-0.5)(-7.96875)}{-8.96875}$

$= -1.05575$

$k=3$: $x_4 = x_3 - \frac{(x_3 - x_2) f(x_3)}{f(x_3) - f(x_2)}$

$= -1.05575 - \frac{[-1.05575-(-1.5)](0.51164)}{0.51164 - (-7.96875)}$

$= -1.05575 - \frac{[0.44425](0.51164)}{8.48039}$

$= -1.08255$

$\vdots$

Since x_7 and x_8 are equal (for the chosen accuracy level), therefore, an estimate of the only (real) root of $f(x)$ is $x^\star \approx -1.10530$.

Advantage of the Secant Method v.s. the Newton Method:

- Does not require computation of the derivatives $f'(x_k)$.

Disadvantages:

- Requires two starting points x_0 and x_1.
- Converges slower than Netwon's method (but still faster than the bisection method).

The iterates x_k in Newton/secant methods need not converge to a root. However, the iterates x_k in the bisection method always converge to a root, but only slowly.

Exercise: Let $f(x) = x^3 - x^2 - x - 1$. For the following, stop computation when the relative error is less than 1%.

1. Use Newton's method to estimate the root of $f(x)$ by letting $x_0 = 2$.
2. Use the Secant method to estimate the root of $f(x)$ by letting $x_0 = 3$ and $x_1 = 2$.

Exercise: Find a root of $f(x) = x^3 - 6x^2 + 11x - 6.1$ by

1. four iterations of the Newton method, with $x_0 = 3.5$.
2. Four iterations of the Secant method with $x_0 = 2.5$ and $x_1 = 3.5$.

Exercise: Estimate a root of $f(x) = 7e^{-x} \sin x - 1$

1. Using Newton's method, letting $x_0 = 0.3$, and executing at least 3 iterations.
2. Using the Secant method, letting $x_0 = 0.5$ and $x_1 = 0.4$, executing at least 3 iterations.
3. Compare the answers from both previous parts by evaluating the function. Which answer is more accurate?